Lakshya JEE 2.0 2025

Maths

DPP: 6

Matrices

Q1 Compute the inverse of the matrix.

$$A=\begin{bmatrix}3 & -1 & 1\\ -15 & 6 & -5\\ 5 & -2 & 5\end{bmatrix}$$

and verify that $A^{-1}A=I_3$.

Q2 Use product $\begin{bmatrix}1 & -1 & 2\\ 0 & 2 & -3\\ 3 & -2 & 4\end{bmatrix}\begin{bmatrix}-2 & 0 & 1\\ 9 & 2 & -3\\ 6 & 1 & -2\end{bmatrix}$ to solve system of equations

$x-y+2z=1$

$2y-3z=1$

$3x-2y+4z=2$

Q3 Solve the following system of equations by matrix method, where $x\neq 0, y\neq 0, z\neq 0$

$\frac{2}{x}-\frac{3}{y}+\frac{3}{z}=10, \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=10, \frac{3}{x}-\frac{1}{y}$.

$+\frac{2}{z}=13$

Q4 If $A^2-A+I=0$, then the inverse of A is

(A) $I-A$

(B) $A-I$

(C) A

(D) $A+I$

Q5 If $\begin{bmatrix}1 & -\tan\theta\\ \tan\theta & 1\end{bmatrix}\begin{bmatrix}1 & \tan\theta\\ -\tan\theta & 1\end{bmatrix}^{-1}$,

$=\begin{bmatrix}a & -b\\ b & a\end{bmatrix}$

then

(A) $a=1, b=1$

(B) $a=\cos 2\theta, b=\sin 2\theta$

(C) $a=\sin 2\theta, b=\cos 2\theta$

(D) None of these

Q6 If a, b and c are the roots of the equation $x^3+2x^2+2x+1=0$ and Δ

$=$

$$\begin{vmatrix}3 & a+b+c & a^3+b^3+c^3\\ a+b+c & a^2+b^2+c^2 & a^4+b^4+c^4\\ a^2+b^2+c^2 & a^3+b^3+c^3 & a^5+b^5+c^5\end{vmatrix}$$

then

(A) $\Delta=6$

(B) $\Delta=0$

(C) $\Delta=-6$

(D) Two values of Δ are possible

Q7 Let $A=\begin{bmatrix}2 & 0 & 7\\ 0 & 1 & 0\\ 1 & -2 & 1\end{bmatrix}$ and

$B=\begin{bmatrix}-x & 14x & 7x\\ 0 & 1 & 0\\ x & -4x & -2x\end{bmatrix}$ are two matrices such that $AB=(AB)^{-1}$ and $AB\neq I$ (where I is an identity matrix). Then Tr

$\left(AB+(AB)^2+(AB)^3+(AB)^4+(AB)^5\right.$

$\left.+(AB)^6\right)$

is equal to $(\mathrm{Tr}(A))$ denotes the trace of matrix A)

(A) 2 (B) 6

(C) 8 (D) 16

Q8 Cayley-Hamilton theorem states that every square matrix satisfies its own characteristic equation. For all

the questions of this passage take, $A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}$

The matrix A satisfies the matrix equation

(A) $A^2-4A-5I=0$

(B) $A^2-4A-5=0$

(C) $A^2+4A-5I=0$

(D) $A^2+4A-5=0$

Q9 Cayley-Hamilton theorem states that every square matrix satisfies its own characteristic equation. For all the questions of this passage take, $A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}$

Which of the following is inverse of A ?

(A) $\frac{1}{5}(A-4I)$

(B) $\frac{1}{5}(A+4I)$

(C) $\frac{1}{4}(A-5I)$

(D) $\frac{1}{4}(A+5I)$

Q10 Which of the following is/are correct?

(A) If A and B are two square matrices of order 3 and A is a non-singular matrix such that $AB=O$, then B must be a null matrix.

(B) It A, B, C are three square matrices of order 2 and $\det(A)=2, \det(B)=3, \det(C)=4$, then the value of $\det(3ABC)$ is 216

(C) It A is a square matrix of order 3 and $\det(A)=\frac{1}{2}$, then $\det\left(\operatorname{adj} A^{-1}\right)$ is 8

(D) Every skew symmetric matrix is singular

Q11 If $A=\begin{pmatrix}1 & 4 & 0\\ 5 & 2 & 6\\ 1 & 7 & 1\end{pmatrix}$ $\& A^{-1}$,

$=\frac{1}{36}\left(\alpha A+\beta A^2+\gamma I\right)$

then $\frac{|\alpha+\beta+\gamma|}{3}$ is equal to P, then sum of digit of P will be (where I is identity matrix)

Q12 Let P and Q are two matrices such that $P^T=3P-4I$ and $Q^T=3Q-I$ then trace of $\operatorname{adj}(\operatorname{adj}(2PQ))$ is_____ (order of P and Q is 3×3)

Q13 If A and B are two matrices of same order 3×3, where $|A|=2$ and $B=\begin{pmatrix}3 & 2 & 5\\ 2 & 3 & 8\\ 7 & 2 & 9\end{pmatrix}$, then match column-I with column-II.

Column-I

Column-II

(A) The value of $|\operatorname{adj}(\operatorname{adj} A)|$ is (P) 2^{16}

(B) The value of $|\operatorname{adj}(AB)|$ is (Q) 16

(C) The value of $|\operatorname{adj}(\operatorname{adj}(2A))|$ is (R) 2304

(D) The value of $|\operatorname{adj}(BA)|$ is (S) 676

(A) $\mathrm{A}\to\mathrm{P};\mathrm{B}\to\mathrm{Q};\mathrm{C}\to\mathrm{R};\mathrm{D}\to\mathrm{S}$

(B) $\mathrm{A}\to\mathrm{Q};\mathrm{B}\to\mathrm{R};\mathrm{C}\to\mathrm{P};\mathrm{D}\to\mathrm{R}$

(C) $\mathrm{A}\to\mathrm{P};\mathrm{B}\to\mathrm{Q};\mathrm{C}\to\mathrm{S};\mathrm{D}\to\mathrm{R}$

(D) $\mathrm{A}\to\mathrm{Q};\mathrm{B}\to\mathrm{P};\mathrm{C}\to\mathrm{S};\mathrm{D}\to\mathrm{R}$

Q14 If $\begin{bmatrix}2 & 1\\ 3 & 2\end{bmatrix}A\begin{bmatrix}-3 & 2\\ 5 & -3\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then A is equal to

(A) $\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}$

(B) $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$

(C) $\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

(D) $\begin{bmatrix}0 & 1\\ 1 & 1\end{bmatrix}$

Q15 If A is an orthogonal matrix, then A^{-1} equals

(A) A (B) A^T

(C) A^2 (D) None of these

Q16 Let A and B are two square matrices of order 2×2, such that $A=\begin{bmatrix}\frac{1}{2} & 1\\ -\frac{1}{2} & \frac{1}{3}\end{bmatrix}$ and $B=I+2A+3A^2+4A^3+\ldots$ upto infinite terms, if $I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then

(A) $b_{11}=\frac{2}{25}$

(B) $b_{12}=2b_{21}$

(C) $b_{22} = \frac{-9}{25}$
(D) $b_{12} = \frac{21}{25}$

Q17 If A is the matrix $\begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix}$ then $A - \frac{1}{3}A^2 + \frac{1}{9}A^3 \ldots + \left(-\frac{1}{3}\right)^n A^{n+1} + \ldots \infty = \frac{3}{13}\begin{bmatrix} 1 & a \\ b & 1 \end{bmatrix}$. Find $\left|\frac{a}{b}\right|$.

Answer Key

Q1	Verified
Q2	x=0, y=5, z=3
Q3	x=1/2, y=1/3, z=1/5
Q4	(A)
Q5	(B)
Q6	(A)
Q7	(B)
Q8	(A)
Q9	(A)
Q10	(A, B)
Q11	2
Q12	48
Q13	(B)
Q14	(A)
Q15	(B)
Q16	(C)
Q17	3

Android App | iOS App | PW Website